PROJECT PRESENTATION

DDoS Attack Detection Using Deep Learning

LSTM | CIC-DDoS2019 | TensorFlow | FastAPI

Why detect DDoS attacks with deep learning

The attack

A DDoS attack floods a service with traffic from many sources until real users are locked out.

The gap

Fixed rules and signatures miss new or mixed attack patterns.

The approach

Learn flow-level traffic patterns and classify each flow as BENIGN or ATTACK.

Data and preprocessing

CIC-DDoS2019: labelled flow CSVs with about 80 numeric features per flow

Step	What happens
Clean	Drop IPs, ports and timestamps; remove inf, NaN and duplicates
Label	BENIGN = 0, any attack family = 1
Split	70 / 15 / 15 stratified train, validation and test
Scale	StandardScaler fitted on training data only, clipped to +/-10
Balance	Class weights; validation and test keep the natural ratio

LSTM model design

Layer	Setting
Input	(1 x F) per flow, or (F x 1) feature sequence
LSTM 1	64 units, returns sequences, dropout 0.3
LSTM 2	32 units, dropout 0.3
Dense	32 units, ReLU
Output	1 unit, sigmoid: attack probability

Adam (0.001), binary cross-entropy, batch 512, early stopping, class weights

Evaluation beyond accuracy

Precision

Share of alerts that are real attacks.

Recall

Share of real attacks that are caught.

Benign recall

Normal traffic correctly passed; BENIGN is the minority class.

Compared with Logistic Regression, Random Forest and an MLP on the same split. **Results: [add LSTM and baseline scores after training on the full dataset]**

Real-time serving with FastAPI

Endpoint	Purpose
GET /health	Status, feature count, layout, threshold
GET /features	Exact feature names the model expects
POST /predict	One flow in; label, probability, latency out
POST /predict_batch	Many flows in; one result per flow

Same scaling, clipping and reshaping as training; missing features return HTTP 422

Limitations and next steps

Limitations

- Lab-generated data may not transfer to other networks
- One time step per flow acts much like a dense network
- Live use needs a flow generator such as CICFlowMeter

Next steps

- Windows of consecutive flows for real sequences
- Multi-class detection of attack families
- Cross-day testing and threshold tuning